



🕒 Time: 5 minutes

🕒 Outcome: Distinguish nominal and real GDP and calculate the GDP deflator.

🕒 Difficulty: Intermediate

Real versus Nominal GDP



THE CORE IDEA

Nominal GDP values current production at current prices, while real GDP values it at base-year prices. Multiply current quantities by the designated base-year prices and add across final goods. GDP is a strong measure of market production for its purpose, but it omits some dimensions of welfare. Compare nominal and real GDP to separate price changes from changes in production. Real-GDP growth equals the change in real GDP divided by the initial real GDP.



HOW TO RECOGNIZE IT

- Nominal GDP uses current-year prices and current-year quantities.
- Real GDP uses base-year prices with current-year quantities.
- A change in nominal GDP can reflect prices, quantities, or both; a change in real GDP reflects quantities.
- Use the GDP deflator to compare nominal GDP with real GDP for the same year.



WATCH OUT

Do not use the new level rather than the initial real-GDP level as the percentage-growth denominator. Real-GDP growth equals the change in real GDP divided by the initial real GDP.



WORKED EXAMPLE: SEPARATING PRICE AND OUTPUT CHANGES

One-good economy: price and output

	Quantity	Price	Nominal GDP	Real GDP
Base year	100	\$5	\$500	\$500
Current year	110	\$6	\$660	\$550

An economy produces 100 units in the base year at \$5 each and 110 units this year at \$6 each. Nominal GDP this year is $110 \times \$6 = \660 . Real GDP at base-year prices is $110 \times \$5 = \550 . The GDP deflator is $(\$660 / \$550) \times 100 = 120$, separating the price increase from the output increase.



CHECK YOURSELF

Is real GDP calculated by simply subtracting the inflation rate from nominal GDP?



READY?

Return to the game and master this concept.